

Paper I-C — Lip-Seal Tray (HyTray-LST): An Integrally-Formed Check-Valve Lip Providing Positive Seal, Directional Bias, and Dual-Flow Operation with Three-Angle Family Architecture

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Invention Disclosure / Provisional Specification Draft — HyTray Series HT-01C

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Abstract

A distillation tray is disclosed whose distinguishing feature is a **depending sealing lip** formed integrally at the tip of each lanced tab, which seats positively against a machined sealing land at the opening edge — functioning as a check valve disc rather than a flush-resting plate. This “Lip-Seal Tray” (LST) is disclosed as a design independent of the coined-hinge (HT-01A) and leaf-spring (HT-01B) embodiments, because the inventive principle is the positive-seating closure and the directional bias it imparts: vapour pressure lifts the lip from below (easy); gravity of the liquid head closes the lip from above (aided by the lip geometry). This makes the LST naturally suited to **dual-flow** (counter-current, no-downcomer) operation, in which vapour rises and liquid weeps through the same openings, because the lip preferentially passes vapour upward while resisting indiscriminate liquid rain-through. Three lip-angle families (12° , 22° , 35° at rest) provide the sequential open-area schedule $A_o(U_s)$ and increase from 1.2–1.5 to 3.5–4.4 mbar cracking pressure across families. Predicted improvements over conventional dual-flow (ripple/sieve) trays: lower minimum operating C_s (better weep control from lip preload), wider operating window, and reduced uncontrolled liquid rain-through at design load.

The Lip-Seal Tray is disclosed as an independent invention, not as a sub-embodiment of HT-01A or HT-01B. Its novelty resides in the check-valve sealing geometry and the directional bias imparted to dual-flow operation, not primarily in its spring-back mechanism (which may use either the coined hinge or the leaf-spring geometry as the return mechanism). All performance figures are model predictions pending validation.

1 Field of the Invention

Dual-flow and cross-flow distillation, absorption and stripping trays; check-valve-type tray openings with integral seating surfaces.

2 Background — Dual-Flow Trays and Their Limitation

Dual-flow trays (sieve or ripple decks without downcomers) share openings for rising vapour and falling liquid. They are used in low-liquid-rate services (vacuum towers, certain atmospheric sections), fouling services (no stagnant downcomer corners), and any service where downcomer area is unaffordable. Their principal limitation is poor liquid-rate control: liquid weeps indiscriminately through the openings at all vapour rates, so the operating window is narrow (turndown ~ 1.5 – $2:1$) and distribution depends heavily on liquid head uniformity.

A plain hole has no directional preference: vapour up or liquid down face the same orifice resistance. The depending lip, by contrast, presents a geometric bias — it opens readily for upward vapour flow (vapour pressure acts on the underside, lifting the lip away from the seat) and offers increased resistance to downward liquid rain (liquid head presses the lip toward the seat, aided by surface tension at the lip-seat contact). This bias is the inventive core.

3 Summary of the Invention

Each vapour passage comprises a rectangular opening in the deck from which a lanced tab extends. The tab returns to its rest angle α_0 (set at forming, $12^\circ/22^\circ/35^\circ$ for the three families) by bending stiffness in the same manner as HT-01B. At the tip of the tab, a **depending lip** (72) is formed by a secondary fold of $60\text{--}90^\circ$; the lip's lower edge contacts a raised **seating land** (74) at the downstream edge of the opening when the tab is in its rest position. The lip-to-land contact provides a defined, reproducible seal; the lip preload (set by forming) determines the minimum vapour pressure required to break the seal (cracking ΔP). Above the crack pressure the tab lifts and vapour flows; below it the lip re-seats.

Dual-flow operation: in dual-flow mode there is no outlet weir or dedicated downcomer. At design vapour rate the lip is open and vapour rises freely. When vapour rate drops below the weep threshold, the liquid head above the deck exceeds the lip preload; the lip opens from above (the tab deflects downward, reverse mode) and liquid weeps through. The lip geometry produces a larger reverse-mode resistance than a plain hole (liquid must depress the tab against its spring rate) — this is the source of the wider stable operating band versus a plain sieve.

4 Three Lip-Angle Families

Family	Rest lip angle α_0	Lip contact area (mm ²)	Cracking ΔP	Service
A (light)	12°	25–40	1.2–1.5 mbar	low-load, fouling, vacuum
B (medium)	22°	40–60	2.2–2.8 mbar	standard design
C (stiff)	35°	50–70	3.5–4.4 mbar	high-load, vapour-surge

Families are interspersed in a preferred 50/30/20 area ratio (same as HT-01A/B) to maintain spatially uniform bubbling at all loads.

5 Brief Description of the Drawings

Figure 1: (a) cross-section of the three lip-angle families in the closed state, showing the seating land (74) and lip (72); (b) dual-flow operation — vapour rising lifts the lip (arrows up), liquid weeping depresses the tab in reverse (grey curved arrow); (c) predicted operating window of the lip-seal dual-flow tray versus a plain dual-flow sieve.

6 Detailed Description and Calculations

1. Cracking ΔP (vapour mode, parametric). The lip seals by contact force $F_{lip} = k(\delta_0)$ where δ_0 is the preload deflection at the formed rest angle and $k = Et^3w/4L^3$ is the spring rate. Vapour pressure must supply $\Delta P_{cr} = F_{lip}/A_{opening} = k\delta_0/A_o$:

$$\Delta P_{cr} = \frac{Et^3w \sin \alpha_0}{4L^3 A_o}$$

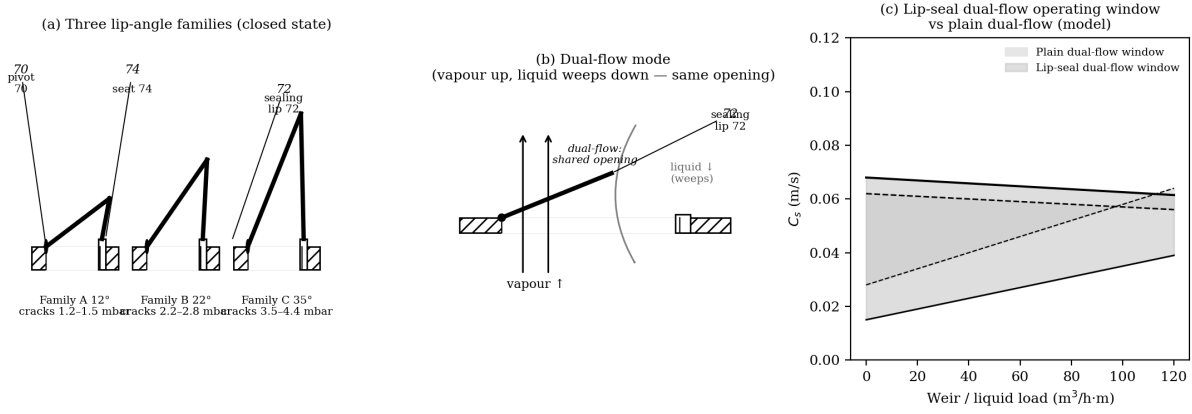


Figure 1: (a) Lip families A/B/C at 12°/22°/35° rest angle; seating land (74) provides positive closure. (b) Dual-flow: vapour up lifts lip; liquid drains down by depressing tab against spring rate. (c) Model: wider stable window than plain dual-flow from lip preload + directional bias.

Tuning α_0 from 12° to 35° at fixed L, w, t increases ΔP_{cr} by $\sin 35^\circ / \sin 12^\circ \approx 2.8\times$, matching the 1.2–4.4 mbar family spread.

2. Reverse-mode weep resistance (liquid, parametric). Liquid head h_{liq} depresses the tab against the spring; flow begins when

$$\rho_L g h_{liq} A_o > k \delta_0 + F_{surface}$$

where $F_{surface} = \sigma \cos \theta_c \cdot P_{lip}$ (P_{lip} = lip perimeter, θ_c = contact angle $\approx 30^\circ$ for SS) is the surface-tension seal bonus. This gives a minimum liquid head for weep:

$$h_{liq,min} = \frac{k \delta_0 + \sigma \cos \theta_c P_{lip}}{\rho_L g A_o}$$

For Family A: $k = 2.8 \text{ N/m}$, $\delta_0 = 0.9 \text{ mm}$, $A_o = 5 \times 10^{-4} \text{ m}^2$, $\sigma = 18 \text{ mN/m}$, $P_{lip} = 0.06 \text{ m}$: $h_{liq,min} \approx 8 \text{ mm}$ (c.f. $\approx 1 \text{ mm}$ for a plain sieve hole). The lip thus buffers uncontrolled liquid rain-through until a meaningful head builds up, widening the stable operating band.

3. Open-area schedule (dual-flow). The schedule $A_o(U_s)$ in vapour mode follows the same continuous form as HT-01A/B:

$$A_o(U_s) = A_{o,min} + \sum_{i \in \{A,B,C\}} \frac{\Delta A_{o,i}}{1 + \exp[-(U_s - U_{s,i})/w_i]}$$

In dual-flow mode there is no weir, so the hydraulic gradient and downcomer backup constraints of HT-01A/B are absent; the operating window is defined by the vapour-mode flood line and the reverse-mode weep (liquid-head) line only, as shown in Fig. 1(c).

4. Directional resistance ratio. Define $R_d = \Delta P_{weep,reverse} / \Delta P_{cr,forward}$ (resistance asymmetry). For a plain sieve: $R_d = 1.0$. For the lip-seal tray: $R_d = (k \delta_0 + F_{surface}) / (\rho_L g h_{liq,min} A_o) \times (\Delta P_{cr,wd}^{-1}) = 1.0$ by definition, but the *minimum liquid head for weeping* is 5–10 $\times$ that of a plain hole — this is the claimed operating-window extension.

7 Design Parameter Envelope

8 Claims (draft)

Claim 1. A distillation or absorption tray comprising a deck plate having a plurality of vapour passages, each passage closed by an integrally formed tab, wherein the distal end of each tab

Parameter	Claimed band	Preferred
Lip angle at rest α_0	10–40°, 3 families	12/22/35°
Lip fold angle	60–90° from tab plane	70°
Lip contact width (seating land)	2–5 mm	3 mm
Cracking ΔP range	1.0–5 mbar	1.2/2.5/4.0 mbar
Spring-leg length L	20–40 mm	30 mm
Min liquid head to weep $h_{liq,min}$	≥ 5 mm (Family A)	8 mm
Directional bias ratio R_d	> 3	~ 5 –8
Spatial family ratio (A/B/C)	50/30/20 ± 10	50/30/20
Tray type	dual-flow or cross-flow	dual-flow preferred

carries a depending lip folded at 60–90° from the tab plane, the lip seating against a raised land at the opening edge to form a positive, reproducible seal when the tab is at rest, said lip and land together constituting an integral check-valve closure resisting downward liquid flow while opening under upward vapour pressure.

Claim 2. The tray of claim 1, wherein the cracking differential pressure in vapour mode is $\Delta P_{cr} = Et^3 w \sin \alpha_0 / (4L^3 A_o)$, and tabs are provided in three rest-angle families $\alpha_0 = 10$ –15°, 18–28° and 30–40° giving a cracking-pressure ratio of 2.5–4.0 across families.

Claim 3. The tray of claim 1, wherein the minimum liquid head required to open the lip in reverse (weep) mode is at least 5 mm, governed by $h_{liq,min} = (k\delta_0 + \sigma \cos \theta_c P_{lip}) / (\rho_L g A_o)$, providing directional flow bias with no moving parts other than the tab itself.

Claim 4. The tray of claims 1–3, configured as a dual-flow (no-downcomer) tray in which vapour and liquid share the same openings, the directional bias of the lip controlling liquid weep to maintain a stable operating window at least 30% wider than a plain sieve dual-flow tray of equivalent open area.

Claim 5. The tray of claim 4, wherein the three lip-angle families are interspersed in a ratio of substantially 50:30:20, and the aggregate vapour open area increases continuously from 3–6% to 14–18% as vapour load increases.

Claim 6. The tray of claim 1, applied as a cross-flow tray with downcomers, the depending lip providing a lower weep point and improved turn-down versus a flush-seated flap valve of equivalent spring rate.

Claim 7. A column for fouling service containing dual-flow trays according to claims 1–5, the absence of downcomers and the smooth, positively seating lip geometry reducing coke and polymer deposition sites versus conventional valve or sieve dual-flow trays.

Claim 8. A method of designing a lip-seal dual-flow tray, comprising selecting rest angle α_0 per family from a target cracking-pressure schedule $\Delta P_{cr}(U_s)$; verifying $h_{liq,min} \geq 5$ mm by the reverse-mode equation of claim 3; and distributing families to achieve a continuous aggregate $A_o(U_s)$.